



Student file tools that run in the browser

PDFs, images, video and audio are processed on the student's device. No account and no file upload are required.

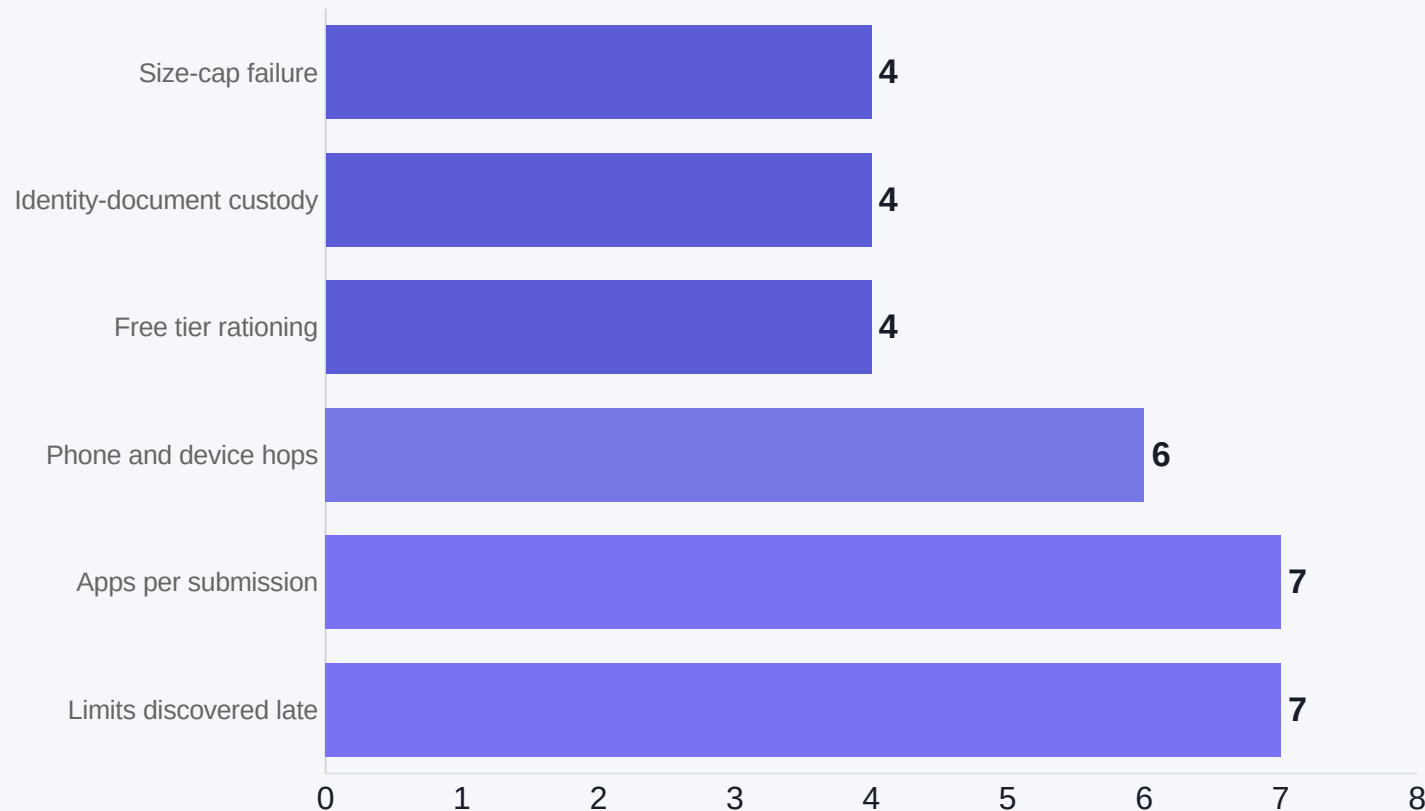
59 tools

15 interviews

25 / 25 tests

Student file work breaks at the final step

Interview evidence from 15 MFU undergraduates across 15 programmes



7 of 15

found the limit only after the work was finished

9 of 15

had a submission rejected because of format or size

The interviews shifted the pitch away from “free” and toward fewer steps, early limits and file custody.

One workflow supports 59 tools

The interface changes settings, while the student follows the same three-step path

1 Select a file

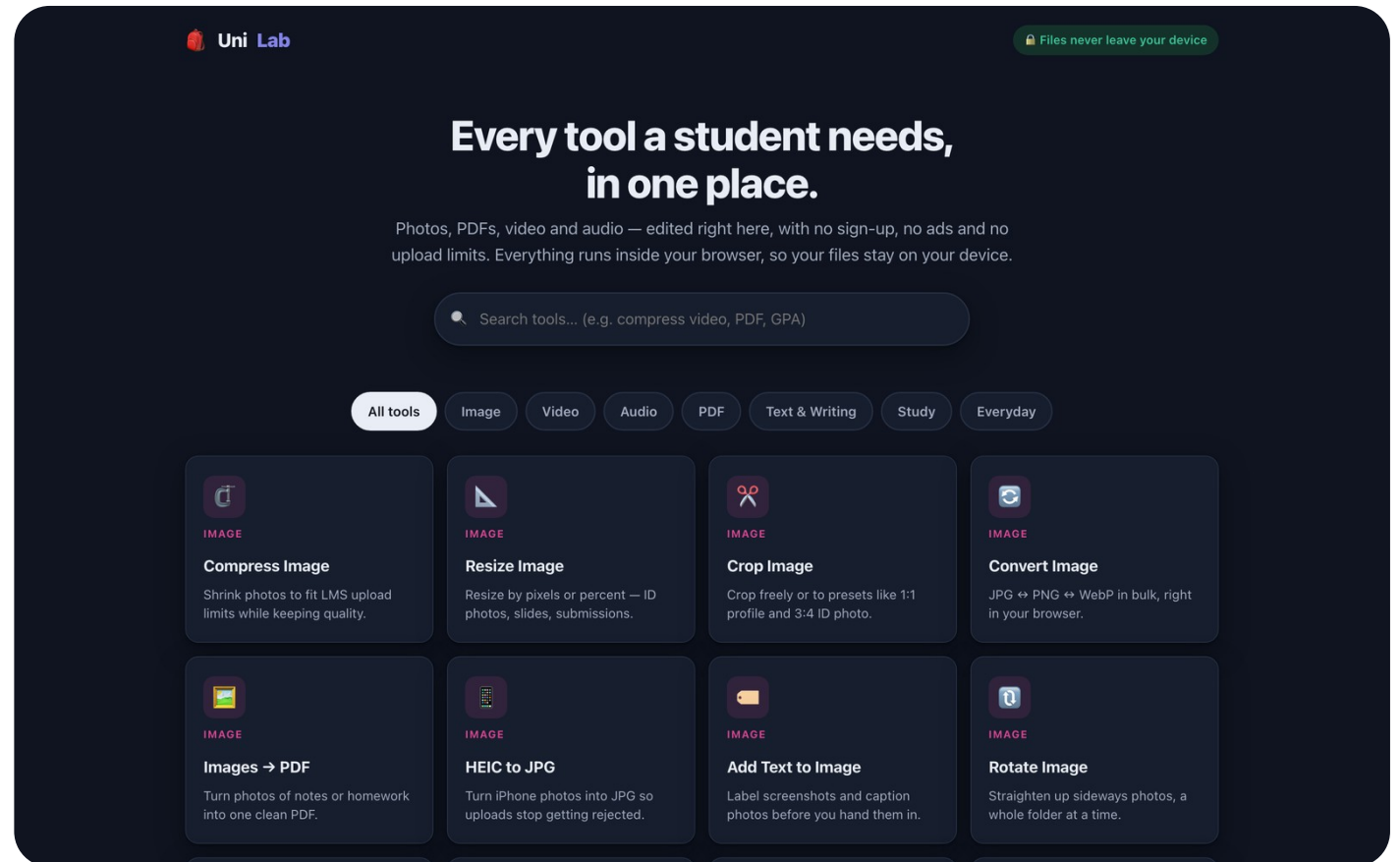
Type and size checks run before the tool starts.

2 Process in the tab

PDF, image and media engines run locally in the browser.

3 Download the result

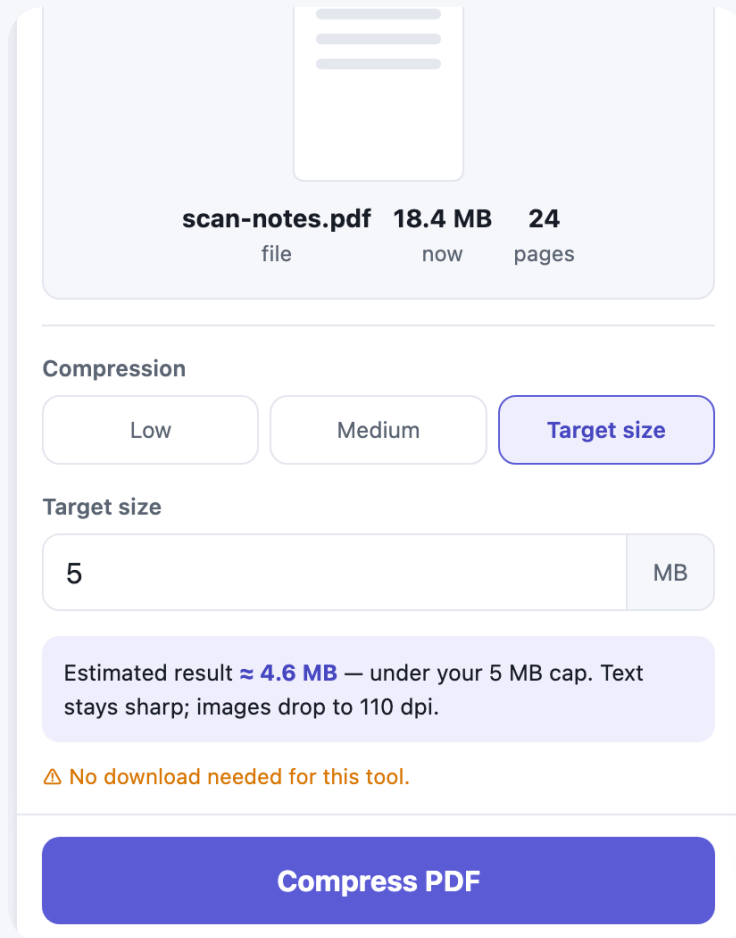
Results stay in memory and expire after 30 minutes.



21 PDF 13 image 11 audio 7 video 2 text 2 study 3 everyday

A size estimate appears before processing

The compress-PDF flow addresses the most common late-stage failure



The screenshot shows a web interface for PDF compression. At the top, a file named 'scan-notes.pdf' is shown with a size of 18.4 MB and 24 pages. Below this, there are three buttons for compression: 'Low', 'Medium', and 'Target size'. The 'Target size' button is selected. Underneath, a 'Target size' input field is set to '5' MB. A light blue box displays the estimated result: 'Estimated result ≈ 4.6 MB — under your 5 MB cap. Text stays sharp; images drop to 110 dpi.' Below this, a small orange triangle icon is followed by the text 'No download needed for this tool.' At the bottom, there is a large blue button labeled 'Compress PDF'.

scan-notes.pdf 18.4 MB 24
file now pages

Compression

Low Medium **Target size**

Target size

5 MB

Estimated result ≈ 4.6 MB — under your 5 MB cap. Text stays sharp; images drop to 110 dpi.

⚠ No download needed for this tool.

Compress PDF

Target-size compression

The student enters the LMS cap. UniLab shows the estimated output size and expected quality before the run.

7 of 15

learned a limit only after completing the work

4 of 15

had to compress after discovering a size cap

6 of 15

worked phone-first and lost time when switching devices

Privacy is enforced in the architecture

Compliance comes from keeping file bytes on the device

0

file uploads per task

No upload endpoint exists in the coursework baseline.

File stays in the browser tab

Engines process the file locally. Optional OCR languages and background-removal models require a disclosed download, while the user's file remains local.

Results live in memory

The vault expires after 30 minutes. Delete now performs the same purge, and tests confirm the bytes become unreachable.

LR1 no file content leaves the device

LR5 memory only and erasable

Current proof and next decisions

Repository checks passed on 23 September 2026

25 / 25

automated tests passing

Build passes

7.40 MB core precache, below the 8 MB budget

59 routes

mounted in the production build

Open evidence

Physical phone benchmark

Complete the pick, process and download flow on a named reference phone.

All-tool processing sweep

Prove no file upload across OCR and media processing jobs.

Course scope decision

Decide whether OCR and saved Workflows remain in course scope.

Instructor feedback

Record the decision before calling the topic and scope approved.

Live product
unilab.ztvmm.live

Coursework evidence
github.com/Thiha-Lynn/unilab